

Emergent Behaviors in City-Scale Evacuation: An Artificial Life Simulation of Multi-Agent Predator-Prey Dynamics

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Abstract—This paper presents an Artificial Life simulation modeling emergent group behaviors during a city-scale evacuation under the threat of external predators. We describe a multi-agent system comprising three distinct agent roles: civilians (prey), soldiers (defenders), and titans (predators) simulated within a dense, bounded urban environment. By defining localized steering rules, sensory perception, and internal physiological states (stress and fatigue), we demonstrate how complex group dynamics emerge without centralized control. Civilians use a pre-computed Dijkstra flow field for evacuation routing while propagating panic dynamically to neighbors. Soldiers coordinate combat maneuvers—rallying to maintain tactical standoff distances and engaging only when local numbers are sufficient. Titans hunt using a pursuit-wander loop. We evaluate the resilience of the crowd, the effectiveness of soldier defense, and the impact of stress and fatigue on the success of the evacuation.

I. INTRODUCTION

The study of collective behaviors and self-organization in animal groups (flocking, herding, and defensive dynamics) represents a core theme in Artificial Life [3]. While simple models like Craig Reynolds’ Boids [1] successfully capture cohesive flocking, modeling populations reacting to active, lethal threats in complex terrains remains a challenge.

Inspired by the narrative framework of popular media (e.g., *Attack on Titan*), this project presents a 3D predator-prey simulation modeling the evacuation of an urban city under attack by giant predators (“Titans”). Rather than relying on centralized control rules, we build the system on individual agents that interact with each other and their environment locally, drawing inspiration from reactive sensory-motor architectures like Braitenberg Vehicles [2]. Large-scale group behaviors—such as structured evacuation streams, herding panic, tactical combat coordination, and crowd-flow disruptions—emerge entirely from these simple local rules.

This report is organized as follows: Section II describes the environment design and navigation graph. Section III covers the behavioral model of civilians, including flocking, stress-fatigue dynamics, and panic contagion. Section IV outlines the tactical states of soldiers, while Section V defines the predator behaviors of Titans. Section VI details the implementation and mathematical formulation. Section VII presents the simulation results, and Section VIII concludes the paper.

II. ENVIRONMENT AND NAVIGATION GRAPH

The simulation is situated in a bounded city environment. The physical layout of the city consists of rectangular building blocks and walls, represented as obstacle volumes, which establish walkable paths, gates, and intersections. Agents must navigate this space while avoiding collisions with structures.

To facilitate pathfinding, the navigation space is modeled as a waypoint graph. Waypoints are scattered throughout the city streets, forming vertices in a network. The navigation system dynamically evaluates lines of sight between waypoints to build links. Agents travel from waypoint to waypoint to mimic destination-oriented movement rather than random walk patterns.

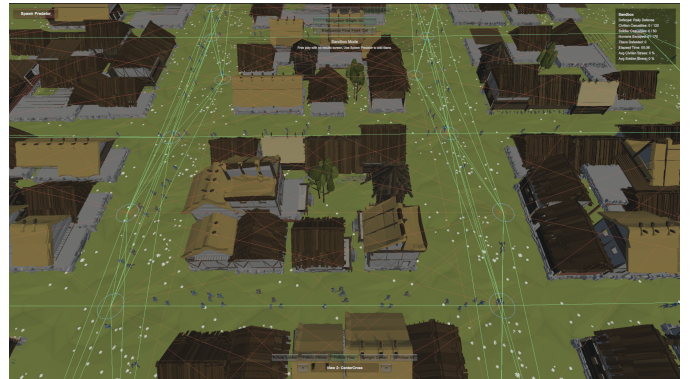


Fig. 1. Visualization of the city navigation graph. Nodes (blue spheres) indicate walkable positions, and links (green/red lines) represent clear and obstructed paths. Buildings and walls are outlined in orange bounds.

During simulation startup, the environment graph is parsed from the Unity hierarchy. Path connectivity is evaluated using line-of-sight checks:

$$\text{Link}(i, j) = \begin{cases} \text{Walkable}, & \text{if } \text{LOS}(\vec{x}_i, \vec{x}_j, r_{\text{clear}}) = \text{True} \\ \text{Blocked}, & \text{otherwise} \end{cases} \quad (1)$$

where r_{clear} is the required agent clearance and LOS represents a segment-sampling check against obstacle volumes.

III. CIVILIAN BEHAVIOR

Civilians (represented as prey agents) are the primary targets of the Titans. They operate under a finite state machine with three main states: *Calm*, *Evacuating*, and *Escaped*.

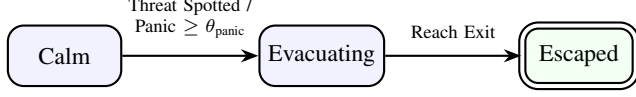


Fig. 2. Prey agent state transition diagram. Calm civilians transition to Evacuating upon encountering threat or through panic contagion. Upon reaching an active evacuation point, they transition to Escaped.

A. Calm Wandering

When no threat is detected, civilians wander the city. Their locomotion is driven by a blend of waypoint-route following and boids-inspired flocking [1]. The steering forces are calculated using separation, alignment, and cohesion:

- **Separation:** Keeps agents from colliding with each other.
- **Alignment:** Aligns the agent's velocity vector with its immediate neighbors.
- **Cohesion:** Pulls the agent toward the center of mass of its local group.

B. Evacuation Pathing & Flow Fields

Normal pathfinding (e.g., A^*) can be computationally expensive when executed by hundreds of agents simultaneously. To optimize evacuation routing, we implement a pre-computed topological Dijkstra flow field solved directly over the sparse waypoint navigation network.

The exit gates of the city are designated as goal nodes, initialized with a cost-to-go of $C(n_{\text{exit}}) = 0$. The waypoint graph is solved in reverse using a Dijkstra shortest-path solver. Each node n in the graph computes its shortest path cost $C(n)$ and stores a pointer to the next-hop node leading toward safety:

$$C(n) = \min_{m \in \text{Neighbors}(n)} \{C(m) + \|\vec{x}_n - \vec{x}_m\|\} \quad (2)$$

$$\text{Next}(n) = \arg \min_{m \in \text{Neighbors}(n)} \{C(m) + \|\vec{x}_n - \vec{x}_m\|\} \quad (3)$$

When an agent enters the evacuating state, it evaluates the optimal entry node n^* on the graph by minimizing its physical distance to the waypoint plus the downstream cost to the exit:

$$n^* = \arg \min_n (\|\vec{x}_{\text{agent}} - \vec{x}_n\| + C(n)) \quad (4)$$

Once this entry node is acquired, the agent navigates the city by traversing the network, updating its target to $n_{k+1} = \text{Next}(n_k)$ upon entering the arrival radius of the current waypoint, until it reaches an exit node ($C(n) = 0$) and escapes.

If a civilian encounters a Titan, they override their graph-based routing with active, high-priority flee steering. Once the threat is evaded, this hybrid formulation allows the agent to immediately re-evaluate its position and resume the evacuation flow field from its new coordinates.

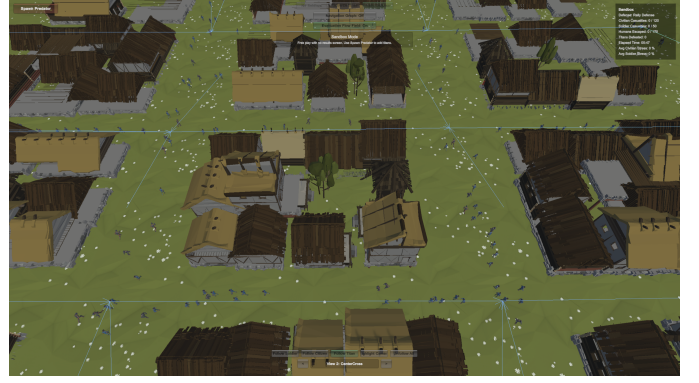


Fig. 3. Vector field visualization of the pre-computed Dijkstra evacuation routes. The arrows point along the shortest path toward the exit gates.

C. Panic Contagion

Civilians do not always need to see a Titan to flee; panic spreads dynamically through the crowd. In our model, panic contagion is a stress-mediated accumulation process rather than an instantaneous binary state transition.

Calm civilians periodically check their neighbors within a contagion radius R_{panic} at a fixed check interval Δt_{panic} . If the number of evacuating civilian neighbors exceeds a minimum count N_{min} , the local panic ratio ρ_i is computed:

$$\rho_i = \frac{|\{j \in N_i \mid s_j = \text{Evacuating}\}|}{|N_i|} \quad (5)$$

where N_i represents the set of civilian neighbors within the contagion radius. This ratio drives a continuous increase in the agent's stress level $S_i \in [0, 1]$:

$$\Delta S_i = \gamma_{\text{panic}} \cdot \rho_i \quad (6)$$

where γ_{panic} is the panic transmission rate. If the civilian's cumulative stress exceeds the evacuation threshold θ_{evac} , they transition to the evacuating state:

$$s_i = \text{Calm} \quad \text{and} \quad S_i \geq \theta_{\text{evac}} \implies s_i \rightarrow \text{Evacuating} \quad (7)$$

This formulation models crowd panic as a cumulative emotional build-up, capturing the behavioral inertia and panic buildup of individuals under threat.

IV. SOLDIER BEHAVIOR

Soldiers are specialized prey agents equipped to defend the city. They patrol the city streets and attempt to slow down or defeat the Titans. Soldiers navigate using five states: *Patrol*, *Alert*, *Rally*, *Engage*, and *Withdraw*.

A. Threat Assessment

Soldiers patrol the waypoint graph in the *Patrol* state. If they detect a Titan or notice a high ratio of fleeing civilians nearby, they transition to *Alert* and immediately clear their current patrol route pauses.

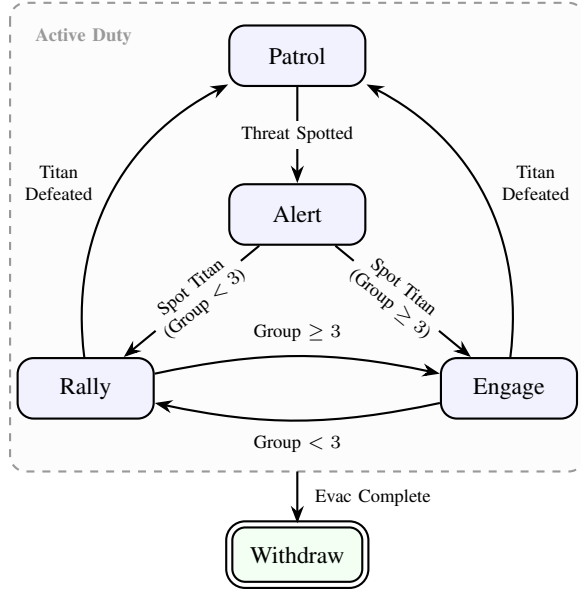


Fig. 4. Soldier agent state transition diagram. Soldiers start in Patrol, transition to Alert upon threat detection, enter the Rally circle if a Titan is sighted, transition to Engage when local soldier numbers are sufficient, and transition to Withdraw once the civilian evacuation is complete.



Fig. 5. Soldiers executing a cooperative rally around a Titan. They maintain a standoff circle, ready to engage once their local numbers are sufficient.

B. Rallying and Engagement

Unlike civilians, soldiers do not flee immediately. However, attacking a Titan alone is lethal. To survive, soldiers execute a group-based tactical model:

- **Rally State:** If a Titan is detected but the number of nearby combat-ready soldiers is below a threshold N_{combat} (typically 3), the soldier circles the Titan. They maintain a standoff distance ($d_{\text{min}}-d_{\text{max}}$, set to 12–28 units), retreating if the Titan gets too close and strafing laterally.
- **Engage State:** Once enough soldiers join the rally, they transition to *Engage*, closing in on the Titan and applying periodic melee attacks.

This grouping behavior prevents soldiers from running in and dying individually, representing a more realistic cooperative defense.

To prevent soldiers from clustering or forming single-file lines during the Rally state, their lateral circling direction is assigned using the parity of their unique agent identifier:

$$\text{Sign}_{\text{strafe}} = (-1)^{\text{ID} \pmod{2}} \quad (8)$$

This breaks local spatial symmetry, splitting the rallying unit into clockwise and counter-clockwise circling sub-groups that naturally envelop the Titan from both sides. When situated within the standoff range ($[d_{\text{min}}, d_{\text{max}}]$), the agent also blends a stabilization force to damp radial oscillations:

$$\vec{f}_{\text{stabilize}} = -\beta_{\text{damping}} \cdot \frac{\vec{v}_{\text{agent}}}{\|\vec{v}_{\text{agent}}\|} \quad (9)$$

where β_{damping} is a stabilization gain.

The decision to transition from Rally to Engage is evaluated over a shared support radius R_{support} around the target Titan's position:

$$N_{\text{combat}} \leq |\{j \in \text{Soldiers} \mid \|\vec{x}_j - \vec{x}_{\text{Titan}}\| \leq R_{\text{support}}\}| \quad (10)$$

This spatial query creates an emergent, consensus-based cooperative engagement. If the local soldier density around the Titan is sufficient, all participating agents advance; if casualties reduce the local count below N_{combat} , they pull back to the safety of the standoff orbit.

C. Withdrawal

Soldiers are not meant to fight indefinitely. Once the civilian population has mostly evacuated or been eliminated, the soldiers withdraw. They transition into the evacuation state and follow the Dijkstra flow field to exit the city. The withdrawal condition is evaluated dynamically based on the ratio of active civilians remaining:

$$N_{\text{civilian_active}} \leq \max(N_{\text{threshold}}, \lambda_{\text{withdraw}} \cdot N_{\text{civilian_total}}) \quad (11)$$

where $N_{\text{threshold}}$ is a hard floor and $\lambda_{\text{withdraw}}$ is the withdrawal ratio (typically 5%). Once met, all soldiers immediately trigger their evacuation routines.

V. STRESS AND FATIGUE MODEL

To capture the breakdown of human movement and tactical effectiveness under threat, we implement a physiological model tracking stress and fatigue.

A. Stress Dynamics

Agents build stress $S_i \in [0, 1]$ based on proximity to Titans, active fleeing, and panic contagion. When actively fleeing a threat, stress increases continuously:

$$\Delta S_{\text{flee}} = \gamma_{\text{flee}} \cdot \Delta t \quad (12)$$

Furthermore, the physical proximity to a Titan increases stress quadratically relative to the threat distance d and perception radius R :

$$\Delta S_{\text{prox}} = \gamma_{\text{prox}} \cdot \left(1 - \text{Clamp}\left(\frac{d}{R}, 0, 1\right)\right) \cdot \Delta t \quad (13)$$

where γ_{flee} and γ_{prox} are stress gain coefficients.

Stress directly couples into the steering weights and agent capabilities to model physiological degradation:

- **Civilians:** Increased stress triggers evacuation behaviour and increases flee weights, but disrupts group cohesion. The flocking and wandering weights are modulated by:

$$w'_{\text{sep}} = w_{\text{sep}} \cdot (1 + S_i \cdot \alpha_{\text{sep}}) \quad (14)$$

$$w'_{\text{coh}} = w_{\text{coh}} \cdot (1 - S_i \cdot \alpha_{\text{coh}}) \quad (15)$$

$$w'_{\text{wander_civ}} = w_{\text{wander_civ}} \cdot (1 + S_i \cdot \alpha_{\text{wander_civ}}) \quad (16)$$

where α_{sep} , α_{coh} , and $\alpha_{\text{wander_civ}}$ are coupling coefficients. This math models how extreme panic reduces group cohesion (pulling toward others) while increasing personal separation (crowd pushing) and wander noise (erratic scattering).

- **Soldiers:** Soldiers build stress slower than civilians. Unlike civilians, stress decreases their random wandering to force tactical path focus:

$$w'_{\text{wander_sol}} = w_{\text{wander_sol}} \cdot (1 - S_i \cdot \alpha_{\text{wander_sol}}) \quad (17)$$

However, high stress degrades combat effectiveness, causing melee damage to scale down:

$$\text{Damage}_{\text{effective}} = \text{Damage}_{\text{base}} \cdot (1 - S_i \cdot \alpha_{\text{damage}}) \quad (18)$$

where α_{damage} represents the stress attack penalty.

B. Fatigue Dynamics

Fatigue accumulates over time when agents are in motion. Standard walking accumulates fatigue at a low rate γ_{moving} , whereas sprinting (active fleeing, evacuating, or combat engagement) accumulates fatigue at a high rate γ_{sprint} :

$$\text{Fatigue}_{t+\Delta t} = \text{Clamp}(\text{Fatigue}_t + \gamma_{\text{fatigue}} \cdot \Delta t, 0, 1) \quad (19)$$

High fatigue applies a penalty to both maximum speed and acceleration. This prevents agents from running at top speed indefinitely and models physical exhaustion:

$$v_{\text{effective}} = v_{\text{target}} \cdot (1 - \lambda_{\text{speed}} \cdot \text{Fatigue}) \quad (20)$$

$$a_{\text{effective}} = a_{\text{target}} \cdot (1 - \lambda_{\text{accel}} \cdot \text{Fatigue}) \quad (21)$$

where λ_{speed} and λ_{accel} are speed and acceleration penalty coefficients. Titans do not experience stress or fatigue, maintaining consistent movement parameters throughout.

VI. TITAN BEHAVIOR

Titans represent the predators in the simulation. They spawn at the outer city gates, representing an invasion force entering from the wild.

Titans wander randomly along the waypoint graph if no prey is detected. When a human enters their detection radius, they switch to pursuit mode, using a predictive trajectory tracking model:

$$\vec{x}_{\text{predict}} = \vec{x}_{\text{prey}} + \vec{v}_{\text{prey}} \cdot \tau_{\text{lead}} \quad (22)$$

where τ_{lead} is the pursuit lead time. Once the Titan is within its attack radius, it eats the prey, pausing momentarily in a celebration state before resuming its hunt. Titans have a health pool and can be defeated by soldiers, in which case they trigger a defeat animation and are destroyed.

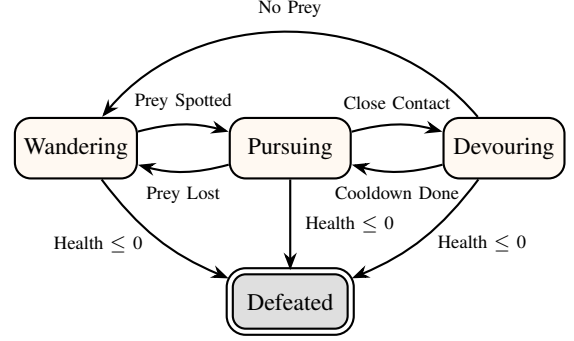


Fig. 6. Titan agent state machine. Titans wander the waypoint graph, acquire and pursue prey within detection range, and devour prey upon close contact.

VII. SIMULATION AND MATHEMATICAL DETAILS

The simulation update loop runs at a fixed timestep to ensure consistent physics evaluation. The steering forces are blended to produce the final acceleration vector for each agent:

$$\vec{a}_{\text{steer}} = \text{Clamp} \left(\frac{\vec{v}_{\text{desired}} - \vec{v}_{\text{current}}}{\Delta t}, a_{\text{max}} \right) \quad (23)$$

For a prey agent, the desired velocity is computed as a weighted sum of steering behaviors:

$$\vec{v}_{\text{desired}} = \sum_k w_k \vec{f}_k \quad (24)$$

where w_k and $\vec{f}_k$ are weights and direction vectors for separation, alignment, cohesion, wandering, path following, obstacle avoidance, boundaries, and threat fleeing. The weights are dynamically modulated by physiological factors (stress and fatigue) as described in Section V.

A. Pathfinding and Graph Queries

While evacuation routing utilizes the global cost-to-go Dijkstra flow field (detailed in Section III), normal calm wandering uses a topological Breadth-First Search (BFS) solver. When a calm agent arrives at the end of its current route, it selects a random target waypoint and builds a topological shortest path:

$$\text{Route} = \text{BFS}(n_{\text{start}}, n_{\text{target}}) \quad (25)$$

Since the waypoint network is sparse and BFS has $O(V + E)$ complexity, this minimizes computational load for non-critical movements.

B. Spatial Hashing

To handle high-frequency spatial queries efficiently, we implement a 2D spatial hash grid in the simulation manager. Proximity checks for flocking, predator detection, and panic contagion are restricted to adjacent cells of size L_{cell} :

$$\text{Cell}(x, z) = (\lfloor x/L_{\text{cell}} \rfloor, \lfloor z/L_{\text{cell}} \rfloor) \quad (26)$$

During the update loop, agents register their positions to the corresponding hash bucket. Querying agents then only

scan their own cell and the 8 adjacent cells. This drops the complexity of proximity queries from $O(N^2)$ to $O(N)$, allowing the simulation to scale to hundreds of agents at real-time framerates.

VIII. EMERGENT SIMULATION RESULTS

The simulation was evaluated inside a city environment with 190 initial prey agents (120 civilians and 70 soldiers) and 10 active Titans. In the case of no soldiers for defense, 120 civilians were simply included, the 70 missing soldiers were not replaced with civilians. There were 3 simulations run per batch to reduce variance.

A. Crowd Evacuation Streams

Without threats, civilians flow smoothly along the waypoint paths toward the exit, forming structured, linear streams. The Dijkstra flow field successfully prevents pathfinding bottlenecks at intersections.

B. Disruption by Titans

When Titans enter the city, they disrupt the crowd streams. Civilians scatter off the main paths to flee, causing local traffic collisions and herding panic. Panic contagion cascades through the neighborhoods, triggering early evacuations.

C. Soldier Rally and Defense

Soldiers successfully coordinate defense. When a Titan enters a street, patrol soldiers alert their nearby peers. They cluster and circle the Titan, maintaining their standoff distance. Once reinforcements arrive, they charge in unison, defeating the Titan. Under this model, soldier casualties are actually unchanged compared to a naive seek-and-attack approach, in our simulation. Naturally, the initial civilian, soldier, and titan values will affect whether increased success from more viable strategies emerges. Thus, this partially explains this results. An additional explanation is provided in part E.

D. Stress and Fatigue Impact

Stress initially increases the evacuation rate due to the panic sprint multiplier. However, as fatigue accumulates, the average speed of the remaining crowd drops. The tail end of the evacuation crowd slows down, making them highly vulnerable to pursuing Titans.

E. Discussion

While these results may initially appear to be divergent from the expected behavior, they can additionally (beyond the hyperparameters of initial population counts) be explained by the clustered spawning of the titans. The rally behavior is meant to allow soldiers to group up against isolated titans, but the fact that the titans all spawn together and that there are 10 of them makes this scenario less likely to occur in our simulations. Nevertheless, there is still increased efficiency to be found as evidenced by the higher civilian survival rate. Moreover, we found that the large map size and patrol ranges for the soldiers meant that they were often not given a chance to group up. Instead, they fled and failed to inflict damage while fleeing from titans, unlike in the naive scenario where they were able to manage this. This was another factor that played a role in the surprising results.

IX. CONCLUSION AND FUTURE WORK

We have presented an Artificial Life simulation of city evacuation dynamics. By combining localized steering, Dijkstra flow fields, panic contagion, group tactical rallying, and physiological models (stress/fatigue), we demonstrate the emergence of complex group-level evacuation behaviors. The rally defense model shows significant defensive improvements over naive solo engagement.

Future work includes adding dynamic obstacle destruction by Titans, implementing building entry/hiding behaviors for civilians, and introducing evolutionary parameter optimization to allow soldiers to evolve better cooperative combat strategies over successive runs.

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TABLE I
SIMULATION EVACUATION STATISTICS

Config	Civilian Survival	Soldier Survival	Titans Defeated
No Soldiers	25.6%	—	0
Naive Defense	75.6%	17.6%	9.7
Rally Defense	79.7%	17.6%	9.7